

ORIGINAL ARTICLE

Olezarsen for Hypertriglyceridemia in Patients at High Cardiovascular Risk

Brian A. Bergmark, M.D., Nicholas A. Marston, M.D., M.P.H., Thomas A. Prohaska, M.D., Ph.D., Veronica J. Alexander, Ph.D., André Zimmerman, M.D., Ph.D., Filipe A. Moura, M.D., Ph.D., Sabina A. Murphy, M.P.H., Erica L. Goodrich, M.S., Shuanglu Zhang, M.P.H., Daniel Gaudet, M.D., Ph.D., Ewa Karwatowska-Prokopczuk, M.D., Ph.D., Sotirios Tsimikas, M.D., Robert P. Giugliano, M.D., and Marc S. Sabatine, M.D., M.P.H., for the Bridge–TIMI 73a Investigators*

ABSTRACT

BACKGROUND

From the TIMI Study Group, Division of Cardiovascular Medicine, Brigham and Women's Hospital, and Harvard Medical School, Boston (B.A.B., N.A.M., A.Z., F.A.M., S.A.M., E.L.G., S.Z., R.P.G., M.S.S.); Ionis Pharmaceuticals, Carlsbad (T.A.P., V.J.A., E.K.-P., S.T.), and the Division of Cardiovascular Medicine, Department of Medicine, University of California, San Diego, La Jolla (S.T.) — both in California; and the Department of Medicine, Université de Montréal and Ecogene-21 Clinical Research Centre, Quebec, QC, Canada (D.G.). Dr. Bergmark can be contacted at bbergmark@bwh.harvard.edu or at the TIMI Study Group, Hale Building for Transformative Medicine, 7th Fl., 60 Fenwood Rd., Boston, MA 02115.

*A complete list of the Bridge–TIMI 73a trial investigators is provided in the Supplementary Appendix, available at NEJM.org.

Drs. Bergmark and Marston contributed equally to this article.

This article was published on April 7, 2024, at NEJM.org.

N Engl J Med 2024;390:1770–80.

DOI: 10.1056/NEJMoa2402309

Copyright © 2024 Massachusetts Medical Society.

Reducing the levels of triglycerides and triglyceride-rich lipoproteins remains an unmet clinical need. Olezarsen is an antisense oligonucleotide targeting messenger RNA for apolipoprotein C-III (APOC3), a genetically validated target for triglyceride lowering.

METHODS

In this phase 2b, randomized, controlled trial, we assigned adults either with moderate hypertriglyceridemia (triglyceride level, 150 to 499 mg per deciliter) and elevated cardiovascular risk or with severe hypertriglyceridemia (triglyceride level, ≥ 500 mg per deciliter) in a 1:1 ratio to either a 50-mg or 80-mg cohort. Patients were then assigned in a 3:1 ratio to receive monthly subcutaneous olezarsen or matching placebo within each cohort. The primary outcome was the percent change in the triglyceride level from baseline to 6 months, reported as the difference between each olezarsen group and placebo. Key secondary outcomes were changes in levels of APOC3, apolipoprotein B, non-high-density lipoprotein (HDL) cholesterol, and low-density lipoprotein (LDL) cholesterol.

RESULTS

A total of 154 patients underwent randomization at 24 sites in North America. The median age of the patients was 62 years, and the median triglyceride level was 241.5 mg per deciliter. The 50-mg and 80-mg doses of olezarsen reduced triglyceride levels by 49.3 percentage points and 53.1 percentage points, respectively, as compared with placebo ($P < 0.001$ for both comparisons). As compared with placebo, each dose of olezarsen also significantly reduced the levels of APOC3, apolipoprotein B, and non-HDL cholesterol, with no significant change in the LDL cholesterol level. The risks of adverse events and serious adverse events were similar in the three groups. Clinically meaningful hepatic, renal, or platelet abnormalities were uncommon, with similar risks in the three groups.

CONCLUSIONS

In patients with predominantly moderate hypertriglyceridemia at elevated cardiovascular risk, olezarsen significantly reduced levels of triglycerides, apolipoprotein B, and non-HDL cholesterol, with no major safety concerns identified. (Funded by Ionis Pharmaceuticals; Bridge–TIMI 73a ClinicalTrials.gov number, NCT05355402.)

REDUCING THE LEVEL OF TRIGLYCERIDE-rich lipoproteins remains an unmet clinical need.¹ Elevated triglyceride-rich lipoprotein and triglyceride levels are associated with heightened cardiovascular risk,¹⁻⁴ and hypertriglyceridemia has direct clinical consequences (e.g., an increased risk of acute pancreatitis), particularly when the condition is severe.¹ Apolipoprotein C-III (APOC3), which is synthesized primarily in the liver and to a lesser extent in the intestine, increases plasma triglyceride levels through several mechanisms, including inhibition of lipoprotein lipase and reduction of hepatic clearance of triglyceride-rich lipoproteins. Loss-of-function variants in *APOC3* are associated with lower triglyceride levels and reduced cardiovascular risk.⁵⁻⁸ Trials of volanesorsen, an unconjugated antisense oligonucleotide targeting *APOC3* messenger RNA (mRNA), in healthy volunteers or patients with chylomicronemia syndromes showed significant reductions in triglyceride levels.⁹⁻¹³ In a recent meta-analysis, volanesorsen reduced the risk of acute pancreatitis events.¹⁴

Olezarsen (ISIS 678354) is an investigational N-acetylgalactosamine-conjugated antisense oligonucleotide targeting *APOC3* mRNA that has been evaluated in a phase 1 study involving healthy volunteers¹⁰ and in a small, phase 2, dose-ranging study.¹⁵ We designed the Bridge-TIMI (Thrombolysis in Myocardial Infarction) 73a trial to assess the efficacy and safety of olezarsen in patients with moderate hypertriglyceridemia (triglyceride level, 150 to 499 mg per deciliter [1.7 to 5.6 mmol per liter]) plus elevated cardiovascular risk and in those with severe hypertriglyceridemia (triglyceride level, ≥ 500 mg per deciliter).

METHODS

TRIAL DESIGN AND OVERSIGHT

In this phase 2b, randomized, placebo-controlled, double-blind trial, we assigned adults who were receiving stable lipid-lowering therapy and had either moderate hypertriglyceridemia plus elevated cardiovascular risk or severe hypertriglyceridemia in a 1:1 ratio to either a 50-mg or 80-mg cohort. Patients were then assigned in a 3:1 ratio to receive monthly subcutaneous olezarsen or matching placebo within each cohort. The objective of this phase 2b trial was to assess the efficacy of these two doses of olezarsen across several lipid measures of clinical importance, while also pro-

viding additional safety data. The randomization ratio between olezarsen and placebo was chosen to provide an adequate number of patients in the placebo group for a statistical comparison, while simultaneously optimizing the number of patients receiving olezarsen for the safety assessment. The treatment period was 12 months, followed by a 13-week follow-up period. All the patients provided written informed consent.

The trial protocol (available with the full text of this article at NEJM.org) was approved by all the relevant ethical oversight committees. The trial was designed in partnership between the members of the TIMI Study Group and Ionis Pharmaceuticals, with data captured at the site level and collected by a contract research organization (Medpace). An independent data and safety monitoring board oversaw safety for the trial. The first draft of the manuscript was written by the first author, and all the authors made the decision to submit the manuscript for publication. The first author vouches for the completeness and accuracy of the data and for the fidelity of the trial to the protocol.

TRIAL POPULATION

Adults (≥ 18 years of age) were eligible for the trial if they had either moderate hypertriglyceridemia plus an increased risk of atherosclerotic cardiovascular disease or severe hypertriglyceridemia. Patients were excluded if they had poorly controlled, severe, or newly recognized diabetes mellitus; a recent acute coronary syndrome, cerebrovascular event, or arterial revascularization; recent acute pancreatitis; or important hepatic or renal laboratory abnormalities. At the time of enrollment, patients were expected to be receiving stable lipid-lowering therapy, according to the local standard of care. Specific criteria for established atherosclerosis or elevated risk, along with full inclusion and exclusion criteria, are provided in the Supplementary Appendix, available at NEJM.org. Qualifying and follow-up laboratory tests were obtained in the fasting state. If a qualifying triglyceride value fell below the required level, two additional tests were allowed, with the average of all values used to determine eligibility for trial enrollment.

OUTCOMES

The primary outcome was the percent change in the triglyceride level from baseline to 6 months, reported as the difference between each olezarsen dose group and placebo. Key secondary outcomes



A Quick Take
is available at
[NEJM.org](https://www.nejm.org)



were changes in other lipid measures, including APOC3, very-low-density lipoprotein (VLDL) cholesterol, total apolipoprotein B, non-high-density lipoprotein (HDL) cholesterol, and low-density lipoprotein (LDL) cholesterol at 6 months; changes in lipid levels at 12 months; the proportion of patients with moderate hypertriglyceridemia at baseline who had a triglyceride level of less than 150 mg per deciliter (the generally accepted clinical threshold for normal reference value) at 6 months and 12 months; and pancreatitis events. A complete listing of secondary and exploratory efficacy outcomes is provided in the statistical analysis plan, available with the protocol.

Assessments of key safety outcomes included monitoring for abnormalities in renal function, platelet counts, and liver enzymes, as well as adverse clinical events. For patients in the 80-mg group, a blinded dose reduction from 80 mg to 50 mg could be considered in patients with a decrease in the platelet count to a level of 50,000 to 100,000 per microliter or for other safety or adverse-event concerns. Blood and urine samples were measured centrally by Medpace Reference Laboratories with the use of commercial assays. Pancreatitis events were to be adjudicated by a central committee in a blinded manner.

STATISTICAL ANALYSIS

We determined that the enrollment of 152 patients would provide the trial with at least 80% power to show a 60% lower triglyceride level for each dose of olezarsen as compared with placebo,^{13,15} assuming a common standard deviation of 73% and 20% withdrawal, with a two-sided alpha level of 0.05. This enrollment would also allow for the performance of general safety assessments. We evaluated the primary outcome according to randomized group among patients who had received at least one dose of either olezarsen or placebo (modified intention-to-treat population). Baseline lipid measurements were defined as the mean of the predose level on day 1 and all levels during the qualification period. Month 6 levels were defined as the mean of week 25 and week 27 levels, and month 12 levels were defined as the average of week 51 and week 53 levels.

We used a washout multiple-imputation approach for missing triglyceride levels.¹⁶ With this approach, data for patients who had discontinued placebo underwent multiple imputation of missing values after the last observed value ac-

cording to levels at the baseline and postbaseline visits. Data for patients who had discontinued olezarsen underwent multiple imputation of missing values according to baseline values and the imputation model for the placebo group. We used the Markov chain Monte Carlo method under the multivariate normality assumption to impute the missing values according to trial group. We used an analysis-of-covariance model (which included log-transformed baseline triglyceride levels) to compare each dose of olezarsen separately with the pooled placebo group in the imputed data sets. We generated least-squares mean estimates of the between-group percent change from baseline from the imputed data sets and combined the data to provide an overall estimate with corresponding 95% confidence intervals and P values.¹⁷ A prespecified testing hierarchy was used to control for multiple comparisons when testing the secondary outcomes, which were tested in hierarchical order up until the first nonsignificant result. The safety population included all the patients who had received at least one dose of olezarsen or placebo. We used a chi-square test to compare safety events or Fisher's exact test if either group had five or fewer events. If a patient in the olezarsen 80-mg group had a dose reduction because of adverse events, that patient would be included in the analysis of the original randomized assignment. All analyses were performed by the TIMI Study Group on an independently held, complete data set, using commercially available statistical software (SAS software, version 9.4).

RESULTS

PATIENTS

Patients were recruited at 24 sites in the United States and Canada from June through September 2022. A total of 154 patients underwent randomization to receive 50 mg of olezarsen (58 patients), 80 mg of olezarsen (57 patients), or placebo (39 patients) once a month. Premature discontinuation of olezarsen or placebo occurred in 14 of 58 patients (24%) in the olezarsen 50-mg group, in 7 of 57 patients (12%) in the olezarsen 80-mg group, and in 3 of 39 patients (8%) in the placebo group (Fig. S1 in the Supplementary Appendix). One patient (2%) in the olezarsen 80-mg group had a dose reduction to 50 mg. In the three groups, 1 patient died (in the olezarsen 50-mg group), 2 were lost to follow-up (1 in each of the olezarsen

Table 1. Characteristics of the Patients at Baseline.*

Characteristic	Olezarsen, 50 mg (N=58)	Olezarsen, 80 mg (N=57)	Placebo (N=39)
Median age (IQR) — yr	63 (54–71)	60 (54–69)	63 (58–71)
Female sex — no. (%)	24 (41)	17 (30)	24 (62)
Race and ethnic group — no. (%)†			
White	54 (93)	52 (91)	35 (90)
Non-Hispanic or Latino	34 (59)	29 (51)	25 (64)
Hispanic or Latino	20 (34)	23 (40)	10 (26)
Black	3 (5)	5 (9)	4 (10)
Non-Hispanic or Latino	2 (3)	5 (9)	1 (3)
Hispanic or Latino	1 (2)	0	3 (8)
Asian	1 (2)	0	0
Median body-mass index (IQR)‡	32.9 (29.7–38.4)	32.4 (27.6–36.6)	33.0 (30.7–37.9)
Medical history — no. (%)			
Pancreatitis	2 (3)	0	0
Atherosclerotic cardiovascular disease	16 (28)	12 (21)	6 (15)
Diabetes mellitus	37 (64)	38 (67)	30 (77)
Chronic kidney disease	8 (14)	9 (16)	5 (13)
Laboratory values			
Triglycerides			
Median value (IQR) — mg/dl	230.0 (182.5–331.5)	241.5 (179.5–357.5)	249.3 (219.5–284.5)
≥500 mg/dl — no. (%)	5 (9)	7 (12)	4 (10)
Median APOC3 (IQR) — mg/dl	15.3 (12.0–19.3)	13.4 (11.5–17.9)	15.8 (13.2–18.7)
Median VLDL cholesterol (IQR) — mg/dl	41.0 (34.0–62.5)	43.0 (31.0–56.7)	41.5 (36.0–61.0)
Median non-HDL cholesterol (IQR) — mg/dl	132.2 (103.5–156.5)	126.0 (110.0–150.5)	134.5 (110.0–161.5)
Median apolipoprotein B (IQR) — mg/dl	90.8 (75.5–105.0)	93.0 (78.0–107.0)	94.0 (78.0–114.5)
Median LDL cholesterol (IQR) — mg/dl	83.8 (59.5–106.0)	80.5 (62.0–102.0)	81.5 (62.0–104.5)
Median glycated hemoglobin (IQR) — %	6.3 (5.9–7.3)	6.4 (5.8–7.0)	6.6 (6.0–7.3)
Median creatinine (IQR) — mg/dl	0.8 (0.7–1.0)	0.9 (0.8–1.1)	0.8 (0.7–0.9)
Median platelet count (IQR) — no./μl	239,000 (198,000–283,000)	237,000 (196,000–288,000)	238,000 (223,000–295,000)
Lipid-lowering therapy — no. (%)			
Any	57 (98)	53 (93)	39 (100)
Statin	49 (84)	46 (81)	32 (82)
Ezetimibe	5 (9)	2 (4)	3 (8)
Fibrate	5 (9)	9 (16)	11 (28)
N-3 fatty acids	12 (21)	14 (25)	8 (21)
Niacin	0	0	1 (3)
PCSK9 inhibitor	3 (5)	1 (2)	1 (3)
≥2 therapies	14 (24)	17 (30)	16 (41)

* To convert the values for cholesterol to millimoles per liter, multiply by 0.02586. To convert the values for creatinine to micromoles per liter, multiply by 88.4. To convert the values for triglycerides to millimoles per liter, multiply by 0.01129. APOC3 denotes apolipoprotein C-III, HDL high-density lipoprotein, IQR interquartile range, LDL low-density lipoprotein, PCSK9 proprotein convertase subtilisin kexin type 9, and VLDL very-low-density lipoprotein.

† Race and ethnic group were reported by the patients.

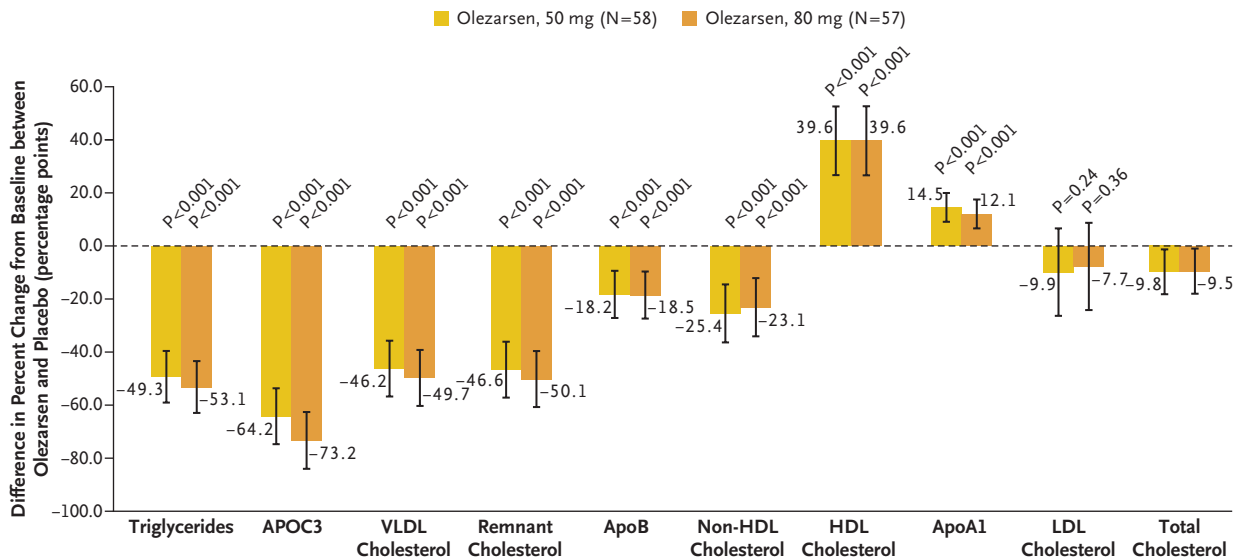
‡ The body-mass index is the weight in kilograms divided by the square of the height in meters.

groups), and 1 withdrew consent (in the olezarsen 50-mg group).

The characteristics of the patients at baseline are provided in Table 1. The median age was 62 years (interquartile range, 55 to 70), 90% of the patients had moderate hypertriglyceridemia, and

42% were women; 92% were White (among whom 38% identified as Hispanic or Latino), and 8% identified as Black. Two thirds of the patients had diabetes, 97% were receiving lipid-lowering therapy, and 31% were receiving two or more lipid-lowering therapies. The median triglyceride level

A Key Lipid End Points at 6 Months as Compared with Placebo



B Triglyceride Levels through 12 Months

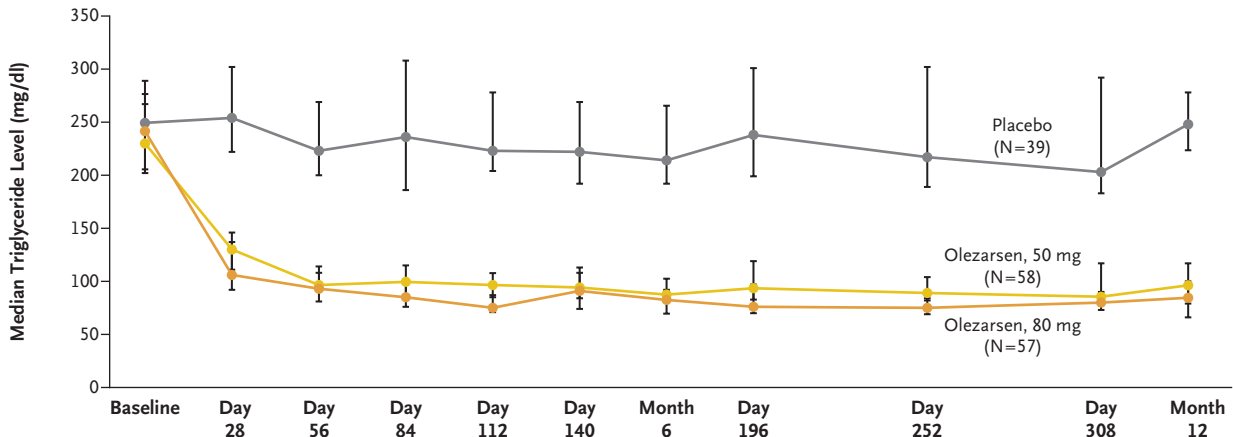


Figure 1. Effect of Olezarsen on Key Lipid Levels at 6 Months and Triglyceride Levels through 12 Months.

Panel A shows the effect of the two doses of olezarsen (50 mg and 80 mg) on key lipid outcomes at 6 months, as compared with placebo. P values are shown for the percent change from baseline for each dose group as compared with the change in the placebo group. Because of the hierarchical order of testing for secondary outcomes, no P values could be reported for the between-group differences in total cholesterol levels. Panel B shows median triglyceride levels from randomization through 12 months. Values at 6 months were calculated as the mean of day 168 and day 182 values. Values at 12 months were calculated as the mean of day 350 and day 364 values. In Panels A and B, I bars indicate 95% confidence intervals. Remnant cholesterol is calculated as the total cholesterol level minus high-density lipoprotein (HDL) and low-density lipoprotein (LDL). To convert the values for cholesterol to millimoles per liter, multiply by 0.02586. To convert the values for triglycerides to millimoles per liter, multiply by 0.01129. ApoA1 denotes apolipoprotein A1, ApoB apolipoprotein B, APOC3 apolipoprotein C-III, and VLDL very-low-density lipoprotein.

was 241.5 mg per deciliter (2.7 mmol per liter; interquartile range, 192 to 324 mg per deciliter [2.2 to 3.7 mmol per liter]). The representativeness of the trial patients relative to the broader population with hypertriglyceridemia is described in Table S1. The baseline characteristics were generally well balanced across the trial groups. However, the proportion of women in the olezarsen groups was lower than that in the placebo group, and fewer patients were taking a fibrate in the olezarsen 50-mg group (9%) than in the placebo group (28%).

OUTCOMES

At 6 months, triglyceride levels were significantly reduced by both doses of olezarsen as compared with placebo. Patients in the placebo group had a mean decrease of 7.8% (95% confidence interval [CI], 0.2 to 15.3) from baseline in the triglyceride level, whereas patients in the olezarsen 50-mg

group had a mean decrease of 57.1% (95% CI, 50.9 to 63.2) and those in the 80-mg group had a mean decrease of 60.9% (95% CI, 54.7 to 67.1). These results led to an absolute difference in the olezarsen groups as compared with the placebo group of 49.3 percentage points (95% CI, 39.5 to 59.0) in the 50-mg group and 53.1 percentage points (95% CI, 43.4 to 62.9) in the 80-mg group ($P<0.001$ for both comparisons) (Fig. 1A and Table S2).

Among 128 patients with moderate hypertriglyceridemia at baseline and nonmissing triglyceride values at 6 months, a triglyceride level of less than 150 mg per deciliter at 6 months was reported in 42 of 49 patients (86%) in the olezarsen 50-mg group and in 42 of 45 patients (93%) in the olezarsen 80-mg group, as compared with 4 of 34 patients (12%) in the placebo group ($P<0.001$ for the comparison of each olezarsen dose with placebo) (Fig. 2). The effect of olezarsen on triglyceride levels was evident as soon as 1 month after the initiation of treatment and persisted through the 12-month follow-up (Fig. 1B).

In the olezarsen groups, the difference with placebo in the percent change from baseline in the APOC3 level was 64.2 percentage points (95% CI, 53.6 to 74.7) in the 50-mg group and 73.2 percentage points (95% CI, 62.5 to 83.9) in the 80-mg group ($P<0.001$ for both comparisons) (Fig. 1A). With respect to additional lipid end points, significant reductions were seen with both olezarsen doses in VLDL cholesterol, remnant cholesterol (calculated as the total cholesterol level minus HDL and LDL cholesterol), non-HDL cholesterol, and apolipoprotein B. There was no significant change in LDL cholesterol with either olezarsen dose as compared with placebo.

The effect of olezarsen on triglyceride levels was consistent across key clinical subgroups, including those based on age, sex, moderate as compared with severe hypertriglyceridemia, body-mass index, diabetes status, and fibrate therapy (Fig. S2). In addition, the degree of triglyceride lowering was consistent in a model that was adjusted for age, sex, race, diabetes status, and fibrate use (Table S3).

SAFETY AND ADVERSE EVENTS

The proportions of patients who had an adverse event during treatment through 28 days after the last dose of olezarsen or placebo were similar in the trial groups (Table 2). Serious adverse events

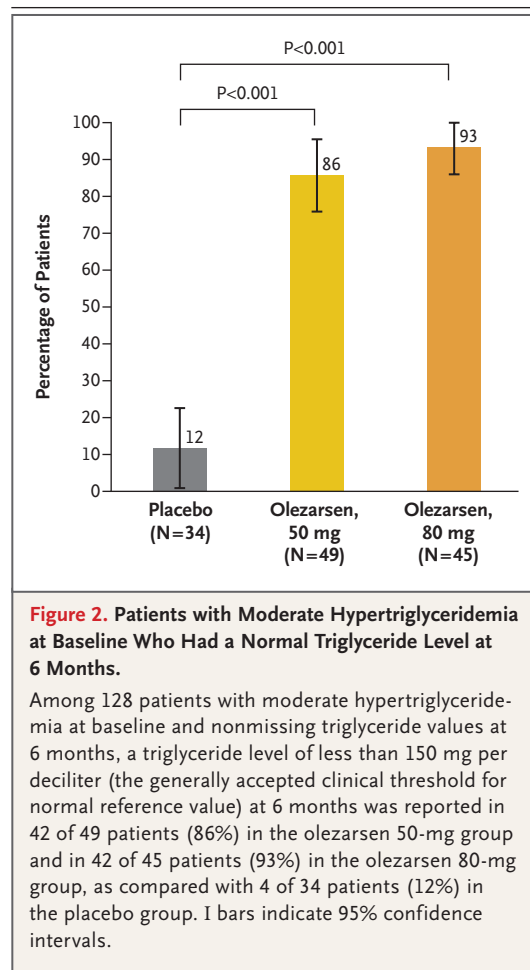


Table 2. Key Safety Measures (Safety Population).*

Safety Measure	Placebo (N=39) no. (%)	Olezarsen, 50 mg (N=58) no. (%)	Risk Ratio vs. Placebo (95% CI)	P Value vs. Placebo	Olezarsen, 80 mg (N=57) no. (%)	Risk Ratio vs. Placebo (95% CI)	P Value vs. Placebo
Adverse event							
Any event	29 (74)	42 (72)	0.97 (0.76–1.24)	0.83	38 (67)	0.90 (0.69–1.16)	0.42
Leading to discontinuation of olezarsen or placebo	0	7 (12)	NA	0.04	3 (5)	NA	0.27
Serious event	2 (5)	4 (7)	1.34 (0.26–6.99)	1.00	7 (12)	2.39 (0.52–10.92)	0.30
Leading to discontinuation of olezarsen or placebo	0	1 (2)	NA	1.00	1 (2)	NA	1.00
Hepatic abnormality†							
Any adverse event	2 (5)	7 (12)	2.35 (0.52–10.74)	0.31	5 (9)	1.71 (0.35–8.37)	0.70
ALT or AST ≥3 times ULN	0	4 (7)	NA	0.15	1 (2)	NA	1.00
ALT ≥ULN							
Any elevation	1 (3)	27 (47)	18.16 (2.57–128.15)	<0.001	21 (37)	14.37 (2.02–102.45)	<0.001
≥3 times ULN	0	4 (7)	NA	0.15	1 (2)	NA	1.00
≥5 times ULN	0	2 (3)	NA	0.51	0	NA	NA
AST ≥ULN							
Any elevation	4 (10)	18 (31)	3.03 (1.11–8.26)	0.025	21 (37)	3.59 (1.34–9.65)	0.004
≥3 times ULN	0	3 (5)	NA	0.27	0	NA	NA
≥5 times ULN	0	1 (2)	NA	1.00	0	NA	NA
Renal abnormality							
Any adverse event	5 (13)	2 (3)	0.27 (0.05–1.32)	0.11	1 (2)	0.14 (0.02–1.13)	0.04
Decrease in eGFR‡							
≥25%	11 (28)	10 (17)	0.61 (0.29–1.30)	0.20	9 (16)	0.56 (0.26–1.22)	0.14
≥30%	8 (21)	6 (10)	0.50 (0.19–1.34)	0.16	4 (7)	0.34 (0.11–1.06)	0.05

Urinary protein:creatinine ratio§							
≥500	10 (26)	8 (14)	0.54 (0.23–1.24)	0.14	8 (14)	0.55 (0.24–1.26)	0.15
≥1000	4 (10)	4 (7)	0.67 (0.18–2.53)	0.71	3 (5)	0.51 (0.12–2.17)	0.44
Platelet count¶¶							
Thrombocytopenia adverse event	1 (3)	0	NA	0.40	2 (4)	1.37 (0.13–14.57)	1.00
Bleeding	2 (5)	3 (5)	1.01 (0.18–5.76)	1.00	3 (5)	1.03 (0.18–5.86)	1.00
Count <140,000/μl	1 (3)	10 (17)	6.72 (0.90–50.44)	0.046	10 (18)	6.84 (0.91–51.31)	0.03
Count <100,000/μl	1 (3)	0	NA	0.40	3 (5)	2.05 (0.22–19.02)	0.64
Other							
Possible hypersensitivity reaction	0	6 (10)	NA	0.08	1 (2)	NA	1.00

§ Data are shown through 28 days after the last dose of olezarsen or placebo. The safety population included all the patients who had received at least one dose of olezarsen or placebo. No pancreatitis events were reported in any group. ALT denotes alanine aminotransferase, AST aspartate aminotransferase, eGFR estimated glomerular filtration rate, NA not applicable, and ULN upper limit of normal range.

¶ No patient had total bilirubin elevation of 2 or more times the ULN. Patients were eligible to enroll in the trial if they had an ALT or AST level up to three times the ULN. An ALT level that was higher than the ULN at baseline occurred in 2 patients (5%) in the placebo group, 6 patients (10%) in the olezarsen 50-mg group, and 4 patients (7%) in olezarsen 80-mg group. An AST level that was higher than the ULN at baseline occurred in 2 patients (5%) in the placebo group, 3 patients (5%) in the olezarsen 50-mg group, and 4 patients (7%) in olezarsen 80-mg group.

¶¶ No patient had a decrease in the eGFR of 50% or more.

§§ Urinary protein was measured in milligrams and urinary creatinine in grams.

|| No patient had a platelet count of less than 75,000 per microliter. At trial entry, a low platelet count was not an exclusion criterion. A baseline platelet count of less than 140,000 per microliter was reported in 1 patient in the placebo group (3%), in no patients in the olezarsen 50-mg group, and in 2 patients (4%) in olezarsen 80-mg group.

The reported event terms listed for possible hypersensitivity reaction included rash, poison ivy rash, lowered blood pressure, asthmatic bronchitis, exacerbation of asthma, injection-site urticaria, and generalized flushing.

* Data are shown through 28 days after the last dose of olezarsen or placebo. The safety population included all the patients who had received at least one dose of olezarsen or placebo. No pancreatitis events were reported in any group. ALT denotes alanine aminotransferase, AST aspartate aminotransferase, eGFR estimated glomerular filtration rate, NA not applicable, and ULN upper limit of normal range.

† No patient had total bilirubin elevation of 2 or more times the ULN. Patients were eligible to enroll in the trial if they had an ALT or AST level up to three times the ULN. An ALT level that was higher than the ULN at baseline occurred in 2 patients (5%) in the placebo group, 6 patients (10%) in the olezarsen 50-mg group, and 4 patients (7%) in olezarsen 80-mg group. An AST level that was higher than the ULN at baseline occurred in 2 patients (5%) in the placebo group, 3 patients (5%) in the olezarsen 50-mg group, and 4 patients (7%) in olezarsen 80-mg group.

‡ No patient had a decrease in the eGFR of 50% or more.

§ Urinary protein was measured in milligrams and urinary creatinine in grams.

¶ No patient had a platelet count of less than 75,000 per microliter. At trial entry, a low platelet count was not an exclusion criterion. A baseline platelet count of less than 140,000 per microliter was reported in 1 patient in the placebo group (3%), in no patients in the olezarsen 50-mg group, and in 2 patients (4%) in olezarsen 80-mg group.

|| The reported event terms listed for possible hypersensitivity reaction included rash, poison ivy rash, lowered blood pressure, asthmatic bronchitis, exacerbation of asthma, injection-site urticaria, and generalized flushing.

were reported in 13 patients (8%); none of these events were considered by the investigator to be related to olezarsen or placebo. Serious adverse events leading to discontinuation occurred in 2 patients in the olezarsen groups — 1 (2%) in the 50-mg group and 1 (2%) in the 80-mg group. A description of serious adverse events is provided in Table S4.

With respect to hepatic safety, there were no cases meeting Hy's law criteria and no major bilirubin elevations (defined as a value at least 2 times the upper limit of the normal range) (Table 2). Any elevation in alanine aminotransferase or aspartate aminotransferase levels was more common with olezarsen than with placebo ($P<0.001$), a difference that was driven by elevations ranging from 1 to less than 3 times the upper limit of the normal range. Elevations of at least 3 times the upper limit of the normal range occurred in no patients in the placebo group, in 4 patients (7%) in the olezarsen 50-mg group ($P=0.15$ as compared with placebo), and in 1 patient (2%) in the olezarsen 80-mg group ($P=1.00$ as compared with placebo).

A decrease in the estimated glomerular filtration rate (eGFR) of at least 30% as compared with baseline occurred in 8 patients (21%) in the placebo group, 6 patients (10%) in the olezarsen 50-mg group ($P=0.16$ as compared with placebo), and in 4 patients (7%) in the olezarsen 80-mg group ($P=0.06$ as compared with placebo) (Table 2). There were no instances of a decrease in the eGFR of at least 50% from baseline in any trial group. A urinary ratio of protein to creatinine of at least 1000 (with protein measured in milligrams and creatinine in grams) was reported in 4 patients (10%) in the placebo group, in 4 patients (7%) in the olezarsen 50-mg group ($P=0.71$ as compared with placebo), and in 3 patients (5%) in the olezarsen 80-mg group ($P=0.44$ as compared with placebo).

A reduction in platelet count to below 100,000 per microliter was observed in 1 patient (3%) in the placebo group, in no patients in the olezarsen 50-mg group ($P=0.40$ as compared with placebo), and in 3 patients (5%) in the olezarsen 80-mg group ($P=0.64$ as compared with placebo) (Table 2). No patients in either dose group had a decrease in the platelet count to below 75,000 per microliter.

Adverse events through the 13-week follow-up after treatment are shown in Table S5. Re-

garding exploratory laboratory testing, there was no significant change in levels of C-reactive protein or glycated hemoglobin in either olezarsen group as compared with placebo at 6 months or 12 months (Figs. S3 and S4).

DISCUSSION

In this phase 2b trial involving patients with hypertriglyceridemia and elevated cardiovascular risk, olezarsen significantly reduced triglyceride levels to a greater degree than can be achieved with currently available therapies. This effect was accompanied by significant reductions in apolipoprotein B and non-HDL cholesterol levels. No major safety concerns were identified.

Hypertriglyceridemia is common, and effective therapies that have an effect on triglyceride-rich lipoproteins are limited.¹ Elevated triglyceride levels put patients at risk for direct clinical consequences, including acute pancreatitis, which is often recurrent when severe hypertriglyceridemia is persistent.^{1,18} In addition, triglyceride-rich lipoproteins appear to have at least the same atherogenic potential per particle as LDL cholesterol.^{1,4,19,20} Observational and mendelian randomization studies have shown strong associations between hypertriglyceridemia, prevalent atherosclerosis, and incident ischemic events across vascular territories.^{4,19,21} Currently available therapies such as statins, ezetimibe, fibrates, and prescription-strength N-3 fatty acids typically lower triglyceride levels by values ranging from less than 10% to approximately 30 to 40%.²² Thus, there is great interest in new therapeutic approaches for reducing circulating levels of triglycerides and triglyceride-rich lipoproteins.

The choice of APOC3 mRNA as a target for triglyceride lowering and cardiovascular risk reduction is supported by genetic and epidemiologic data, as well as the results of early-phase trials. Carriers of loss-of-function variants in APOC3 have been observed to have lower triglyceride levels and a lower risk of subclinical or overt atherosclerosis than noncarriers.⁵⁻⁷ In previous randomized trials, volanesorsen (an unconjugated antisense oligonucleotide against APOC3) reduced triglyceride levels in patients with familial chylomicronemia syndrome (FCS)^{9,11,12} or non-FCS with moderate or severe hypertriglyceridemia.^{11,13} In a meta-analysis of three trials, the use of volanesorsen led to a lower risk of hepatic steatosis²³

and a lower incidence of acute pancreatitis than with placebo.¹⁴ However, despite promising efficacy results, thrombocytopenia was identified as a potential safety concern with volanesorsen.⁹

Olezarsen is a hepatically directed antisense oligonucleotide targeting APOC3. In phase 1–2a trials involving healthy adults¹⁰ and patients with moderate hypertriglyceridemia,^{10,15} the monthly administration of olezarsen at doses of up to 60 mg significantly reduced triglyceride levels. The efficacy and safety of olezarsen at a higher dose and in broader populations has not previously been studied.

In this context, the findings of the Bridge–TIMI 73a trial are encouraging for this therapeutic pathway. The two olezarsen doses reduced triglyceride levels to less than 150 mg per deciliter at 6 months in more than 85% of patients who had moderate hypertriglyceridemia at baseline. There were no pancreatitis events during the treatment period, which precludes assessment of olezarsen efficacy for this clinical end point in this trial. The effect of olezarsen on acute pancreatitis events was also assessed in patients with FCS (as reported now in the *Journal*²⁴) and is also being studied in larger ongoing trials (ClinicalTrials.gov numbers, NCT05610280, NCT05079919, and NCT05552326), including among patients with severe hypertriglyceridemia.

With respect to differences between the two olezarsen doses, although the effects on triglyceride levels were similar at 6 months, the results support a larger reduction with the 80-mg dose at 12 months. Further trials will inform to what extent the higher dose provides additional efficacy for triglyceride lowering and other lipid measures, especially in patients with severe hypertriglyceridemia.

The effect of olezarsen on lipid measures beyond triglyceride levels deserves consideration, especially since previous therapies that have specifically targeted triglycerides have rarely shown cardiovascular benefit. Most recently, the placebo-controlled PROMINENT trial of pemafibrate in patients with mild or moderate hypertriglyceridemia and diabetes showed no material between-group difference in adverse cardiovascular events over a median follow-up of 3.4 years.²⁵ However, pemafibrate decreased APOC3 levels by 28% and triglyceride levels by 26% and increased apolipoprotein B levels by 5%, with no significant change in non-HDL cholesterol. Conversely, in our cur-

rent trial, olezarsen significantly decreased atherogenic particles as reflected by apolipoprotein B and non-HDL cholesterol levels. However, it is not fully understood how this approach to APOC3 reduction affects the metabolism of LDL cholesterol and particles relative to other therapies targeting this or related pathways.¹ Additional trial data, including from a coronary computed tomographic angiography substudy of an ongoing phase 3 trial (NCT05610280), will help to further define the potential for olezarsen to modify cardiovascular risk. Along with the reduction in atherogenic risk markers that was shown in our current trial, these plaque-imaging findings will need to be considered to understand whether a dedicated cardiovascular outcomes trial is justified.

No serious adverse events attributed to olezarsen were reported during treatment. Clinical studies of volanesorsen identified reversible platelet-count abnormalities as a potential concern,^{9–11,15} and in a recent trial of a small interfering RNA targeting APOC3 mRNA, frequent liver-enzyme elevations were observed.²⁶ These measures were monitored closely during the current trial. Although olezarsen was associated with a greater incidence of any elevation in aminotransferase levels, clinically meaningful increases in liver aminotransferases were infrequent, with no significant differences among trial groups, and there were no major elevations in bilirubin. Severe thrombocytopenia occurred in no patients during treatment with olezarsen. Ongoing trials, including two phase 3 trials with hepatic magnetic resonance imaging substudies (NCT05079919 and NCT05552326) should shed further light on safety.

It is important to interpret these results in the context of a changing landscape for metabolic disease management. Antisense oligonucleotide, small interfering RNA, and monoclonal antibody agents targeting the same or related pathways, such as angiopoietin-like protein 3,¹ have been or are in development. It will be critical to understand how agent- and class-level efficacy and safety compare with that of olezarsen while also considering additional features, such as frequency of administration. In addition, lifestyle modifications and pharmacologic therapy for weight loss are evolving. Although the degree of triglyceride lowering with olezarsen was substantially greater than that seen with a glucagon-like peptide 1

(GLP-1) receptor agonist,²⁷ the benefit of APOC3 reduction on cardiovascular outcomes remains to be shown.

As for limitations, in addition to the small size of the trial, we did not evaluate efficacy and safety beyond 1 year after the initiation of treatment. However, open-label extension trials of olezarsen (NCT05681351 and NCT05130450) are under way to gather longer-term data. Moreover, these findings cannot necessarily be applied to patients with specific genetic syndromes or secondary causes of hypertriglyceridemia.

In patients with predominantly moderate hy-

pertriglyceridemia and elevated cardiovascular risk, the monthly administration of olezarsen at a dose of 50 mg or 80 mg significantly reduced triglyceride levels as compared with placebo without major safety concerns. Unlike many therapies targeting triglyceride-rich lipoproteins, olezarsen led to meaningful reductions in levels of apolipoprotein B and non-HDL cholesterol, markers of atherogenic risk.

Supported by Ionis Pharmaceuticals.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

A data sharing statement provided by the authors is available with the full text of this article at NEJM.org.

REFERENCES

- Ginsberg HN, Packard CJ, Chapman MJ, et al. Triglyceride-rich lipoproteins and their remnants: metabolic insights, role in atherosclerotic cardiovascular disease, and emerging therapeutic strategies: a consensus statement from the European Atherosclerosis Society. *Eur Heart J* 2021;42:4791-806.
- Marston NA, Giugliano RP, Im K, et al. Association between triglyceride lowering and reduction of cardiovascular risk across multiple lipid-lowering therapeutic classes: a systematic review and meta-regression analysis of randomized controlled trials. *Circulation* 2019;140:1308-17.
- Khera AV, Won HH, Peloso GM, et al. Association of rare and common variation in the lipoprotein lipase gene with coronary artery disease. *JAMA* 2017;317:937-46.
- Nordestgaard BG, Varbo A. Triglycerides and cardiovascular disease. *Lancet* 2014;384:626-35.
- Jørgensen AB, Frikke-Schmidt R, Nordestgaard BG, Tybjaerg-Hansen A. Loss-of-function mutations in APOC3 and risk of ischemic vascular disease. *N Engl J Med* 2014;371:32-41.
- Pollin TI, Damcott CM, Shen H, et al. A null mutation in human APOC3 confers a favorable plasma lipid profile and apparent cardioprotection. *Science* 2008;322:1702-5.
- Crosby J, Peloso GM, Auer PL, et al. Loss-of-function mutations in APOC3, triglycerides, and coronary disease. *N Engl J Med* 2014;371:22-31.
- Johansen CT, Kathiresan S, Hegele RA. Genetic determinants of plasma triglycerides. *J Lipid Res* 2011;52:189-206.
- Witztum JL, Gaudet D, Freedman SD, et al. Volanesorsen and triglyceride levels in familial chylomicronemia syndrome. *N Engl J Med* 2019;381:531-42.
- Alexander VJ, Xia S, Hurh E, et al. N-acetyl galactosamine-conjugated antisense drug to APOC3 mRNA, triglycerides and atherogenic lipoprotein levels. *Eur Heart J* 2019;40:2785-96.
- Gouni-Berthold I, Alexander VJ, Yang Q, et al. Efficacy and safety of volanesorsen in patients with multifactorial chylomicronaemia (COMPASS): a multicentre, double-blind, randomised, placebo-controlled, phase 3 trial. *Lancet Diabetes Endocrinol* 2021;9:264-75.
- Gaudet D, Brisson D, Tremblay K, et al. Targeting APOC3 in the familial chylomicronemia syndrome. *N Engl J Med* 2014;371:2200-6.
- Gaudet D, Alexander VJ, Baker BF, et al. Antisense inhibition of apolipoprotein C-III in patients with hypertriglyceridemia. *N Engl J Med* 2015;373:438-47.
- Alexander VJ, Karwowska-Prokopczuk E, Prohaska TA, et al. Volanesorsen to prevent acute pancreatitis in hypertriglyceridemia. *N Engl J Med* 2024;390:476-7.
- Tardif JC, Karwowska-Prokopczuk E, Amour ES, et al. Apolipoprotein C-III reduction in subjects with moderate hypertriglyceridaemia and at high cardiovascular risk. *Eur Heart J* 2022;43:1401-12.
- Carpenter JR, Roger JH, Kenward MG. Analysis of longitudinal trials with protocol deviation: a framework for relevant, accessible assumptions, and inference via multiple imputation. *J Biopharm Stat* 2013;23:1352-71.
- Little RJ, Rubin DB. Statistical analysis with missing data. 3rd ed. New York: John Wiley 2019.
- Hansen SEJ, Varbo A, Nordestgaard BG, Langsted A. Hypertriglyceridemia-associated pancreatitis: new concepts and potential mechanisms. *Clin Chem* 2023;69:1132-44.
- Nordestgaard BG. Triglyceride-rich lipoproteins and atherosclerotic cardiovascular disease: new insights from epidemiology, genetics, and biology. *Circ Res* 2016;118:547-63.
- Björnsön E, Adiels M, Taskinen MR, et al. Triglyceride-rich lipoprotein remnants, low-density lipoproteins, and risk of coronary heart disease: a UK Biobank study. *Eur Heart J* 2023;44:4186-95.
- Ference BA, Kastelein JJP, Ray KK, et al. Association of triglyceride-lowering LPL variants and LDL-C-lowering LDLR variants with risk of coronary heart disease. *JAMA* 2019;321:364-73.
- Mach F, Baigent C, Catapano AL, et al. 2019 ESC/EAS guidelines for the management of dyslipidaemias: lipid modification to reduce cardiovascular risk. *Atherosclerosis* 2019;290:140-205.
- Prohaska TA, Alexander VJ, Karwowska-Prokopczuk E, et al. APOC3 inhibition with volanesorsen reduces hepatic steatosis in patients with severe hypertriglyceridemia. *J Clin Lipidol* 2023;17:406-11.
- Stroes ESG, Alexander VJ, Karwowska-Prokopczuk E, et al. Olezarsen, acute pancreatitis, and familial chylomicronemia syndrome. *N Engl J Med* 2024;390:1781-92.
- Das Pradhan A, Glynn RJ, Fruchart JC, et al. Triglyceride lowering with pemafibrate to reduce cardiovascular risk. *N Engl J Med* 2022;387:1923-34.
- Gaudet D, Clifton P, Sullivan D, et al. RNA interference therapy targeting apolipoprotein C-III in hypertriglyceridemia. *NEJM Evid* 2023;2(12). DOI: 10.1056/EVIDoa2200325.
- Lincoff AM, Brown-Frandsen K, Colhoun HM, et al. Semaglutide and cardiovascular outcomes in obesity without diabetes. *N Engl J Med* 2023;389:2221-32.

Copyright © 2024 Massachusetts Medical Society.